Small-Scale Proxies for Large-Scale Transformer Training Instabilities

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https://github.com/fotisk07/DisTrain-project.git

Abstract

We reproduce and investigate two training instabilities described in Wortsman et al. [2023]: *attention logit growth* and *output logit divergence*. We show that both phenomena can be elicited in small Transformers (under 10 M parameters) by training at high learning rates, and that the mitigations proposed by the authors—**qk-layernorm** and **z-loss** regularization—are equally effective at small scale. Our experiments validate the paper's central claim that small models serve as reliable proxies for studying large-scale instabilities, without requiring massive compute.

1 Introduction

Scaling Transformer-based models has driven much of the recent progress in deep learning, from large language models to vision systems. Yet larger models do not always train smoothly: practitioners have repeatedly reported training instabilities that only appear at scale, making them costly and difficult to study Chowdhery et al. [2022], Dehghani et al. [2023]. A team at a large lab may only discover a failure mode after burning weeks of compute on a multi-billion-parameter run, with little insight into the root cause.

Wortsman et al. [2023] propose a practical remedy: *small models can serve as proxies for large-scale instabilities*. Their key observation is that the same failure modes that appear in models with tens of billions of parameters can be reproduced in models with just a few million parameters, simply by training at unusually high learning rates. This makes the instabilities cheap to study, ablate, and fix.

The paper identifies two concrete instabilities and their mitigations:

- **Attention logit growth** Dehghani et al. [2023]: the queries and keys in self-attention layers grow unboundedly, causing the softmax to collapse to a one-hot distribution and learning to stall. The proposed fix is *qk-layernorm*, which applies LayerNorm to queries and keys before the dot product.
- **Output logit divergence** Chowdhery et al. [2022]: the output logits drift to very negative values, decoupling them from the log probabilities. The proposed fix is an auxiliary *z-loss* term that penalises $\log^2 Z$, keeping the softmax partition function near 1.

To summarise the sensitivity of a model to learning rate across all these conditions, the authors introduce a scalar metric called *LR sensitivity*, which measures the expected deviation from optimal loss when sweeping the learning rate over three orders of magnitude.

Our contribution. In this report we independently attempt to reproduce both instabilities at small scale and verify that the proposed mitigations work as claimed. We manage to reproduce one of the two central paper claims of the paper. Our experiments require only modest compute and constitute a self-contained verification of the paper's central claim.

Preprint. Under review.

2 Paper Summary

Wortsman et al. argue that large-scale Transformer training instabilities are not caused by scale per se, but by the interaction between architecture, optimizer, and learning rate. As a result, the same failure modes can be elicited in small models at high learning rates, making them cheap to study and fix. To quantify this, the authors introduce *LR sensitivity*, defined as:

$$\mathcal{S} = \mathbb{E}_{\eta \in [a,b]} \left[\min\big(\ell(\mathcal{A}(\eta)),\, \ell_0\big) - \ell^* \right], \tag{1}$$

where $\ell(\mathcal{A}(\eta))$ is the validation loss when training with learning rate η, $\ell^* = \min_\eta \ell(\mathcal{A}(\eta))$ is the best achievable loss over the swept range, and ℓ_0 is the loss at initialisation (which caps diverged runs). A high $\mathcal{S}$ means the model requires careful LR tuning and is likely to diverge if η is even slightly too large.

Attention logit growth. In self-attention, the logits $z_{ij} = \langle q_i, k_j \rangle / \sqrt{d_h}$ depend on the *product* of query and key vectors. Because both q_i and k_j grow linearly with their respective weight matrix norms, the logit magnitudes grow *quadratically* with parameter norm — much faster than any other feature in the network, which plays a central role in the instability. Once logits become very large, the softmax saturates into a near-one-hot distribution: every token attends to a single fixed position regardless of content, and the model stops learning. First reported in a 22 B parameter ViT Dehghani et al. [2023], the paper shows this also occurs in 9 M parameter models at high LR. The mitigation is **qk-layernorm**: apply LayerNorm to q and k before the dot product, which pins their norms and breaks the quadratic dependence.

Output logit divergence. The model's output logits $y \in \mathbb{R}^V$ produce token probabilities $p_i = e^{y_i}/Z$ where $Z = \sum_j e^{y_j}$. Without weight decay, the logits drift collectively towards very negative values late in training. As $y_i \to -\infty$, $Z \to 0$ and the log-probabilities $\log p_i = y_i - \log Z$ decouple from the logits, causing the loss to stall. First reported in PaLM Chowdhery et al. [2022], the paper reproduces it in a 2.4 M parameter model. The mitigation is a **z-loss** auxiliary penalty $\alpha \log^2 Z$ (with $\alpha = 10^{-4}$), which softly keeps $\log Z \approx 0$ throughout training.

3 Claims Under Verification

Claim 1. *Attention logit growth appears in small Transformers at high learning rates, and qk-layernorm resolves it at both small and large scale, measurably reducing LR sensitivity.* We verify this by training models with and without qk-layernorm across several learning rates, tracking the max attention logit over training, and computing LR sensitivity for each configuration (Section 3.1.1 and Figures 1–2 of Wortsman et al. [2023]). We investigate the origin of this instability by explicitly tracking the norms of the query and key projections. We also perform experiments on shuffled and IID datasets, showing that the same divergence behavior occurs regardless of data structure, thereby supporting the claim that this instability is primarily driven by optimizer dynamics rather than data properties.

Claim 2. *Output logit divergence occurs in Transformer models when weight decay is absent, and z-loss prevents this instability.* We evaluate this claim by training models under the four configurations $\{\text{z-loss, no z-loss}\} \times \{\text{weight decay, no weight decay}\}$ and tracking the mean and variance of the output logits throughout training. Contrary to the findings of Wortsman et al. [2023], we do not observe clear logit divergence in the absence of weight decay.

4 Experimental Setup

All experiments are run with the same decoder-only Transformer training code based on the nanoGPT implementation(**?**) and configured through Hydra. Our goal is not to match the largest models studied in the original paper, but to reproduce the same failure modes in a compact setting that is cheap to run and easy to ablate.

Model. We closely follow the paper's implementation : The model uses rotary positional embeddings, no dropout, bias-free linear layers, and no weight tying between the token embedding matrix and the output projection.

Data and optimisation. Due to size constraints we chose to train on the OpenWebText-style dataset provided in our pipeline, using the standard next-token prediction objective and not on the C4 dataset. The default batch size is 32 sequences of length 256, with one gradient accumulation step.

Training is performed in bfloat16 when available, with `torch.compile` enabled. We use AdamW with $\beta_1 = 0.9$ and $\beta_2 = 0.95$ and decoupled (independent) weight decay as defined in their paper.

5 Results

5.1 Claim 1

Tracking Attention Logits To reproduce Claim 1, we monitor the **maximum attention logits** in the first Transformer layer. This choice is motivated their empirical observations showing that early layers exhibit the largest logits and are the first to diverge. We compare two configurations: with `qk-layernorm` and without `qk-layernorm`, and for each setting we run the experiments with learning rate $\eta \in \{0.1,\ 0.03,\ 0.01,\ 0.003\}$

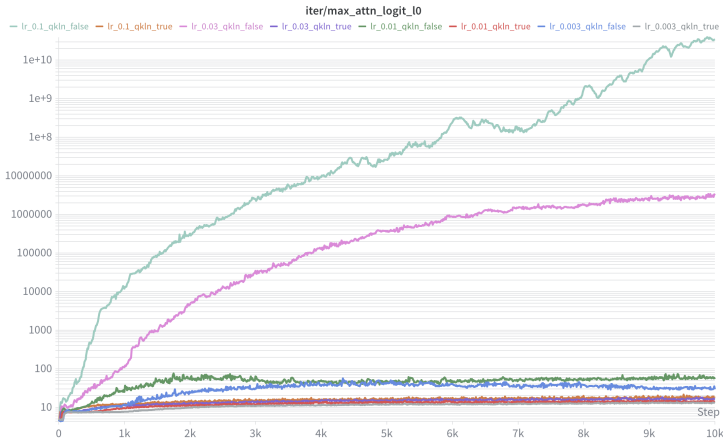
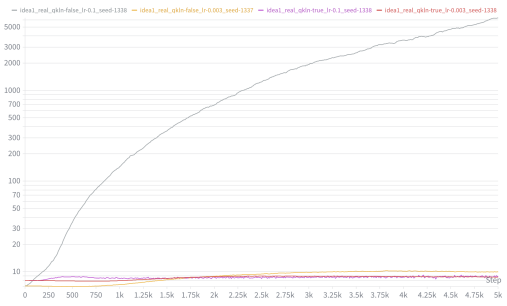


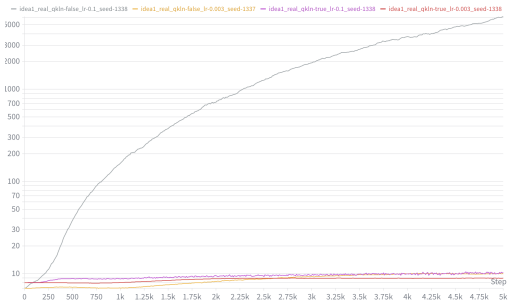
Figure 1: Evolution of the maximum attention logits in the first layer. Divergence occurs at high learning rates when qk-layernorm is not used.

We observe that attention logits diverge when: $\eta \geq 0.03$ and qk-layernorm is disabled. And this behavior is consistent with their findings. We then report the following learning rate sensitivy : **0.025** with qk-layernorm and **1.37** without, and (unfortunately it is not a curve depending, only for a given parameters size, here ≈50M), which follows their results.

Origin of the Instability: Norm Explosion To confirm the origin of this divergence, we track the mean norms of queries and keys, and observe that the norms of Q and K increase rapidly, reproducing the results proposed in appendix of the paper.



(a) Mean norm of Q



(b) Mean norm of K

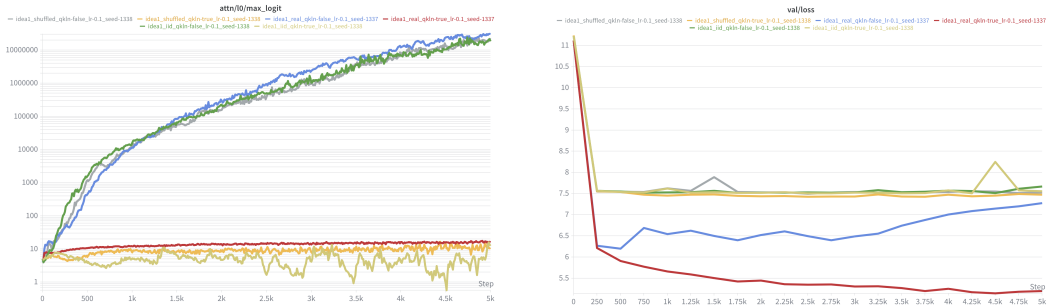
Figure 2: Evolution of query and key norms in the first attention layer.

Optimizer vs Data: Robustness Analysis In theory, for gradient descent, to avoid divergence, one requires with η the learning rate :

$$\eta < \frac{2}{\lambda_{\max}(H)} \text{ and where } H \text{ is the Hessian of the loss: } H_{ij} = \frac{\partial^2 \mathcal{L}}{\partial \theta_i \partial \theta_j}$$

This quantity is too large to compute in practice, but can be approximated. Moreover, H depends on the parameters, but also on the training process and therefore on the data $\mathcal{D}$: $H = H(\theta_{\text{step}}, \mathcal{D})$ where the dynamic is included in θ_{step} (as noted by the authors, the curvature varies during training). The *data-driven* aspect can be interpreted as defining the global loss landscape, while the *optimization-driven* aspect determines where we are located in this landscape and how we evolve within it.

The original work suggests that this instability is optimizer-driven rather than data-dependent. We validate this hypothesis by comparing different dataset constructions. A dataset is said to be IID (independent and identically distributed) if: $x_1, x_2, \ldots, x_n \sim \mathcal{D}$ where all samples are independent and drawn from the same distribution. In practice, this removes any sequential or structural dependencies. A shuffled dataset consists of the same samples but randomly permuted. Unlike IID data, local dependencies may still exist, but global ordering is destroyed.



(a) Maximum attention logits

(b) Validation loss

Figure 3: Comparison across standard, shuffled, and IID datasets. Note that for iid and shuffled it is higher as the model can only overfit the training data without learning to generalize.

As we observe that logit divergence occurs in all settings, our experiments confirm that instabilities are more driven by the dynamics:

$$\text{stability} \sim \frac{\eta}{\lambda_{\max}(H(\theta_{\text{step}}, \mathcal{D}))} \approx \frac{\eta}{\lambda_{\max}(H(\theta_{\text{step}}))}.$$

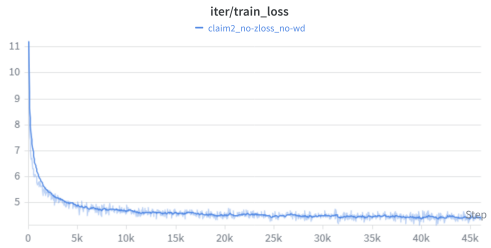
5.2 Claim 2

Our reproduction of Claim 2 was ultimately unsuccessful. While the paper reports a clear output-logit divergence when weight decay is removed, we did not observe the same phenomenon reliably in our small-scale setting. Across the runs we carried out, the mean output logit either remained comparatively well-behaved, or training became unstable in a more global and less interpretable way before any clean divergence pattern emerged.

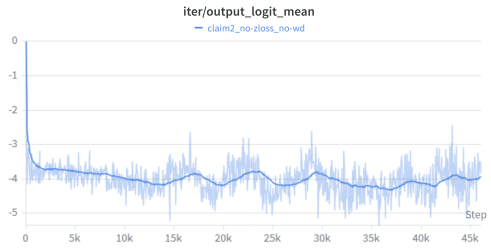
To make the instability easier to trigger, we deliberately explored more aggressive training settings than our default setup. In particular, we disabled gradient clipping, increased the learning rate, removed learning-rate warmup, and disabled learning-rate decay. These changes were intended to stress the optimisation dynamics and reproduce the failure mode described in the paper more quickly. However, this did not lead to a convincing confirmation of the claim.

Instead, we observed two broad outcomes. In some runs, training remained stable enough that the output logit mean did not exhibit the expected runaway behaviour, even in the no-weight-decay configuration. In other runs, optimisation became unstable more abruptly, with large training pathologies that made the behaviour difficult to attribute specifically to the output-logit mechanism discussed by Wortsman et al. [2023]. As a result, our experiments do not provide clear evidence that removing weight decay alone causes the output logit mean to explode in our setting, nor do they allow us to cleanly isolate the stabilising effect of z-loss.

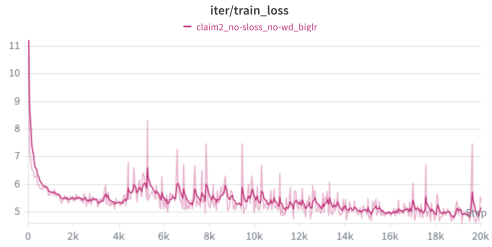
We nevertheless report representative figures from these runs. They illustrate the main outcome of our reproduction attempt: depending on the configuration, we observed either no clear divergence in the output logit mean or severe overall optimisation instability, but not the clean qualitative pattern described in the original paper.



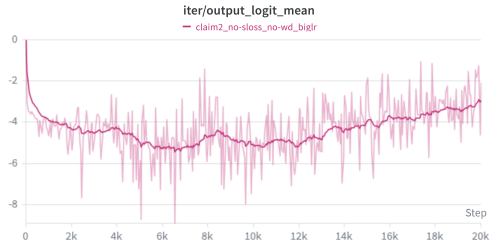
(a) Stable — train loss



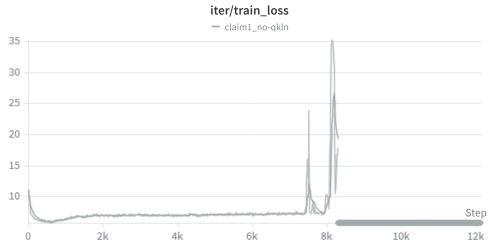
(b) Stable — logit mean



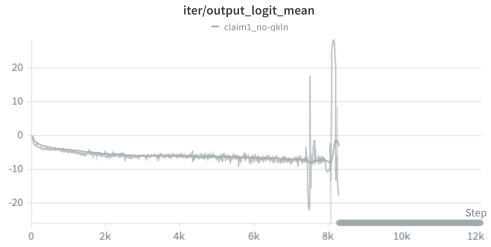
(c) Moderately stable — train loss



(d) Moderately stable — logit mean



(e) Unstable — train loss



(f) Unstable — logit mean

Figure 4: Claim 2 experiments across three regimes. Top: stable training run(lr=0.1). Middle: moderately stable regime with mild fluctuations(lr linear warmup to 0.5). Bottom: unstable regime showing divergence. In three cases we did not observe a collapse to $-\infty$

Takeaway. Our experiments therefore remain inconclusive for Claim 2. They suggest that reproducing output-logit divergence may be more sensitive to implementation details, model scale, or optimisation choices than suggested by the simplified small-scale protocol alone.

References

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